

<https://www.kaggle.com/datasets/khushipitroda/stock-market-historical-data-of-top-10-companies?resource=download>

Which company has the largest change in average annual return over the past 3-5 years?
How did their volatility (standard deviation of daily returns) change before vs. after key events (COVID-19)?

1. Problem Domain & Dataset

This project focuses on the **stock market performance** of leading technology and consumer companies listed on the NASDAQ.

The dataset, "*Stock Market Historical Data of Top 10 Companies*" from Kaggle (sourced via Nasdaq.com), contains **daily stock prices and trading volumes** for major firms such as Apple, Microsoft, Tesla, Amazon, Meta, Netflix, Starbucks, Cisco Systems, Qualcomm, and AMD.

2. Guiding questions

I. Which company experienced the largest change in its **average annual return** over the last 3–5 years?

- This question aims to uncover which of the top 10 NASDAQ companies demonstrated the **strongest growth or decline in returns** when comparing early and recent years in the dataset

II. How did each company's **volatility** (daily return variability) change before and after the onset of the COVID-19 pandemic?

- This investigates how major market events, like COVID-19, influenced investor behavior and price stability in leading NASDAQ companies.

3. How we are going to answer/research our questions

Question i

- We plan to use SQL to calculate each company's **average annual closing price** and then compare those averages across several years.
- Our goal is to identify **which of the top NASDAQ companies** shows the biggest increase or decrease in returns over time.
- This will involve using **aggregations and subqueries** to measure percentage changes in performance.
- The purpose is to understand **market trends and investor shifts** among major tech and consumer firms, while applying SQL techniques that connect directly to real-world financial data analysis.

Question ii

- We plan to compute **daily returns** for each company and then measure their **volatility** before and after March 2020 using SQL functions like `STDDEV()` and conditional logic.
- This will help us compare how stable or risky these companies became following a major global event.
- The analysis is valuable because it connects **data-driven SQL techniques** with **real-world market behavior**, showing how external shocks like COVID-19 affect investor confidence and stock stability.
- It also demonstrates how we can use SQL not just for data storage, but for **quantitative analysis and insight generation**.